Aaron Tran

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Education

University of Waterloo Waterloo, ON

Bachelor of Computer Science *Expected Apr. 2030*

- Relevant Coursework: Designing Functional Programs (CS 135), Algorithm Design & Data Abstraction (CS 136), Algebra (MATH 135), Linear Algebra (MATH 136), Calculus 1 (MATH 137)

Experience & Leadership

Information Communication Technology Intern Toronto, ON

Toronto Catholic District School Board *Mar. 2025 – May 2025*

- Developed an automated file-renaming program in Python to convert VEX Robotics file formats to STL for 3D modeling and editing workflows, making conversions 15.6x faster than the manual process
- Edited video content and collaborated closely with the ICT department head at St. John Paul II, who led the school's Samsung Solve for Tomorrow initiative, helping the team place in the top 12 regional finalists out of 1,000+ schools nationally in Canada

Control Systems Lead

FIRST Robotics Competition Team (High School) *Sep. 2025 – Mar. 2026*

- Led the control systems subteam of a FIRST Robotics Competition team to the provincial championship
- Developed the robot's control systems in C++ and Java using WPILib, making driver-input response 1.5x faster

Projects

LTN (Laser Tag Now) § | *Swift, ARKit, Nearby Interaction, Network.framework* 2026

2nd Place out of 35 teams – SummerHacks

- Built a peer-to-peer iOS laser tag game using Nearby Interaction (Ultra-Wideband) and Network.framework for direct phone-to-phone real-time gameplay, with no backend server or venue Wi-Fi required
- Engineered a UWB aiming system combining ranging with ARKit camera bearing to resolve hits within a 9° cone
- Built a Bonjour and direct-TCP peer-to-peer layer with lobby discovery and a Codable protocol for 8 players

Dowey § | *Swift, SwiftUI, AppKit, Core Animation* 2026

- Built a native macOS window-snapping utility in Swift: hold the Globe key, flick the cursor, and the frontmost window snaps into one of eight zones via the Accessibility API, with multi-monitor and full-screen support
- Designed direction-based zone math with unequal angle widths tuned to flick accuracy as a unit-tested geometry module, plus six live-previewed ring designs in a SwiftUI settings gallery
- Engineered for near-zero overhead: no polling or dependencies, a 10.5 MB idle footprint at 0.00029% CPU, and a Core Animation rewrite that cut interrupt wakeups 2.2x

MarryMap § | *Next.js, React, TypeScript, Supabase, Chrome Extension* 2026

1st Place out of 50+ teams – Cursor Hackathon Toronto

- Built an in-app Vendor CRM tracking every vendor through a full sales pipeline (Lead → Contacted → Quoted → Booked) with quote and deposit amounts, due dates, a per-vendor communication log, and CSV export
- Shipped a collaborative dashboard where couples swipe on vendor options with automatic matching, shareable invites, and per-vendor quote and email history
- Built a Chrome MV3 extension (Supabase/Postgres) that lets guests clip vendors into the couple's CRM

FaceRay § | *Python, OpenCV, MediaPipe, Tauri, Rust, TypeScript* 2026

- Built a real-time AI virtual camera that captures webcam video with OpenCV, tracks a 468-point 3D face mesh via MediaPipe, and outputs to a native OS virtual camera for Discord, Zoom, and Meet
- Implemented gaze correction by anchoring iris landmarks to an attention vector with per-eye temporal smoothing, plus face-confined skin smoothing, anonymization, and background blur
- Architected a Tauri 2.0 desktop shell with a strict data/control-plane split: the Python CV core runs as an isolated sidecar so frames never cross IPC, while a Rust/TypeScript control panel toggles features over a stdio JSON protocol

Technical Skills

Languages: TypeScript, Python, C++, Java, Rust, Swift, JavaScript, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, Tauri, Tailwind CSS, shadcn/ui, ARKit, Network.framework
Tools & Platforms: Git, MongoDB, Supabase, Vercel, Vultr
AI / ML: OpenCV, MediaPipe